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# 4D COVERAGE COLLAPSE ANALYSIS REPORT
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Date: February 05, 2026
Experiment: 4D Boolean Minimization Coverage Modeling

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## MODEL DESCRIPTION
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This report analyzes the coverage collapse behavior of 4D cluster formation in Boolean function minimization across different minterm densities (0.3, 0.5, 0.7, 0.9).

The model predicts when adding more variables (n) stops improving coverage and begins producing redundant clusters.

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## MATHEMATICAL MODEL
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$$\text{Coverage}(n) = (C_{19} \times r^k \times I\_sat) / (\text{density} \times 2^n) \times 100$$

where:

- $C_{19}$ : Number of 4D clusters at n=19
- r : Geometric growth ratio (cluster increase rate)
- k : Counter (k = n - 19, starts at 1 for n=20)
- I_sat : Saturation information density (minterms/cluster)
- n : Number of variables

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## PARAMETERS BY DENSITY
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Density 0.3:
$C_{19}$ = 244 clusters
r = 2.1186
I_sat = 5.92 minterms/cluster

Density 0.5:
$C_{19}$ = 388 clusters
r = 2.2748
I_sat = 9.05 minterms/cluster

Density 0.7:
$C_{19}$ = 845 clusters
r = 2.2432
I_sat = 16.05 minterms/cluster

Density 0.9:
$C_{19}$ = 4755 clusters
r = 2.1834
I_sat = 33.60 minterms/cluster

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Analysis follows on subsequent pages, one per density.
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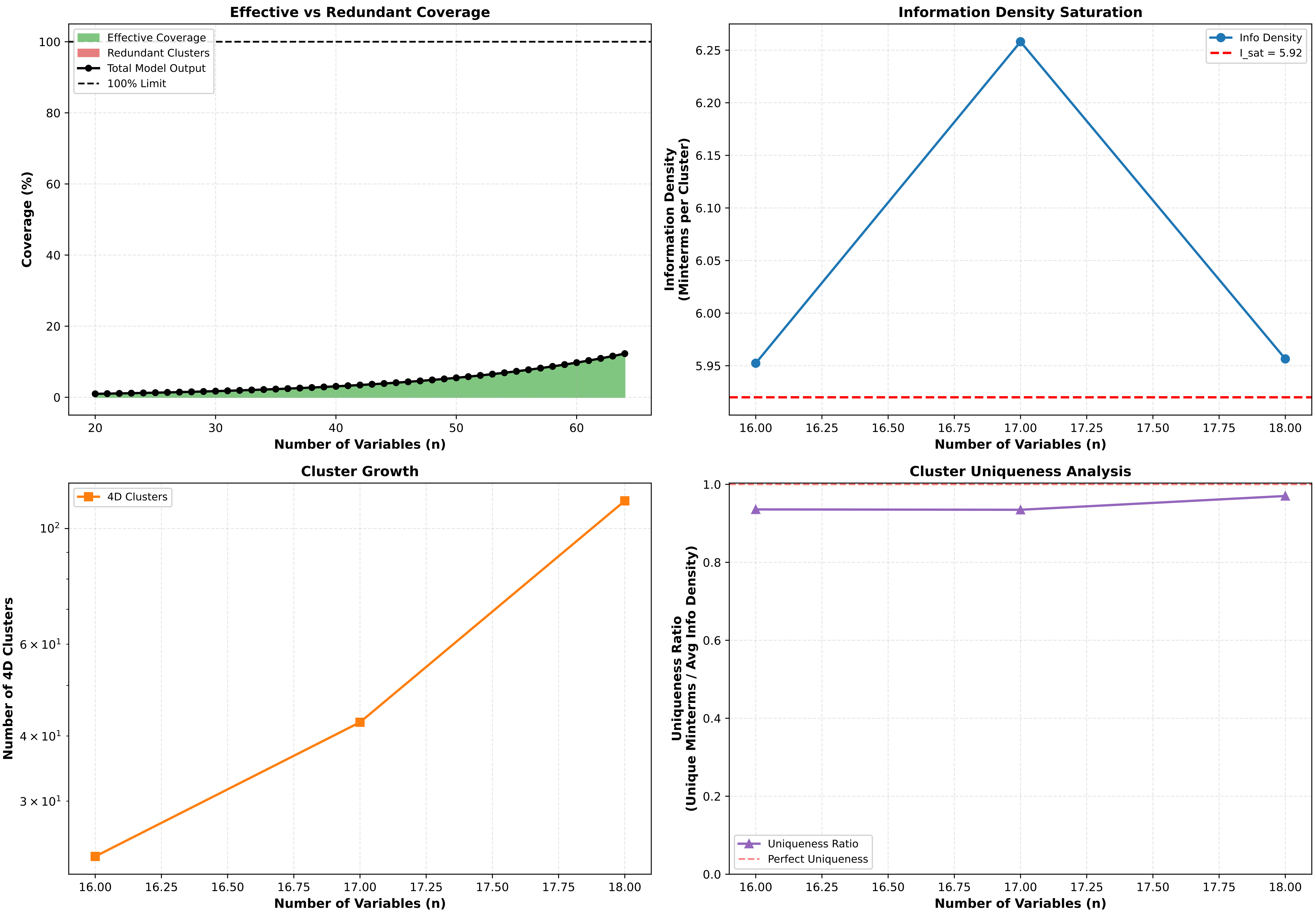
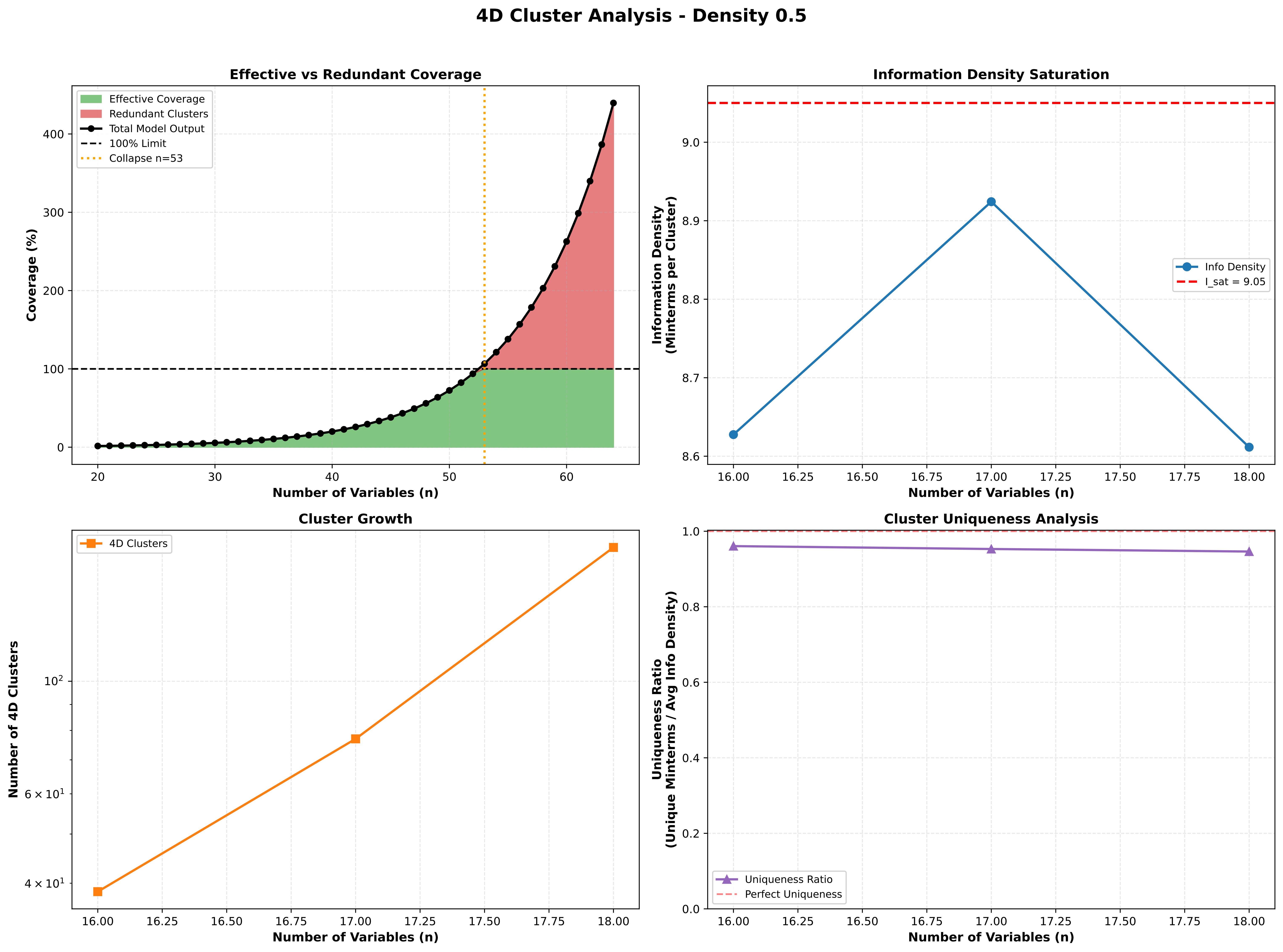
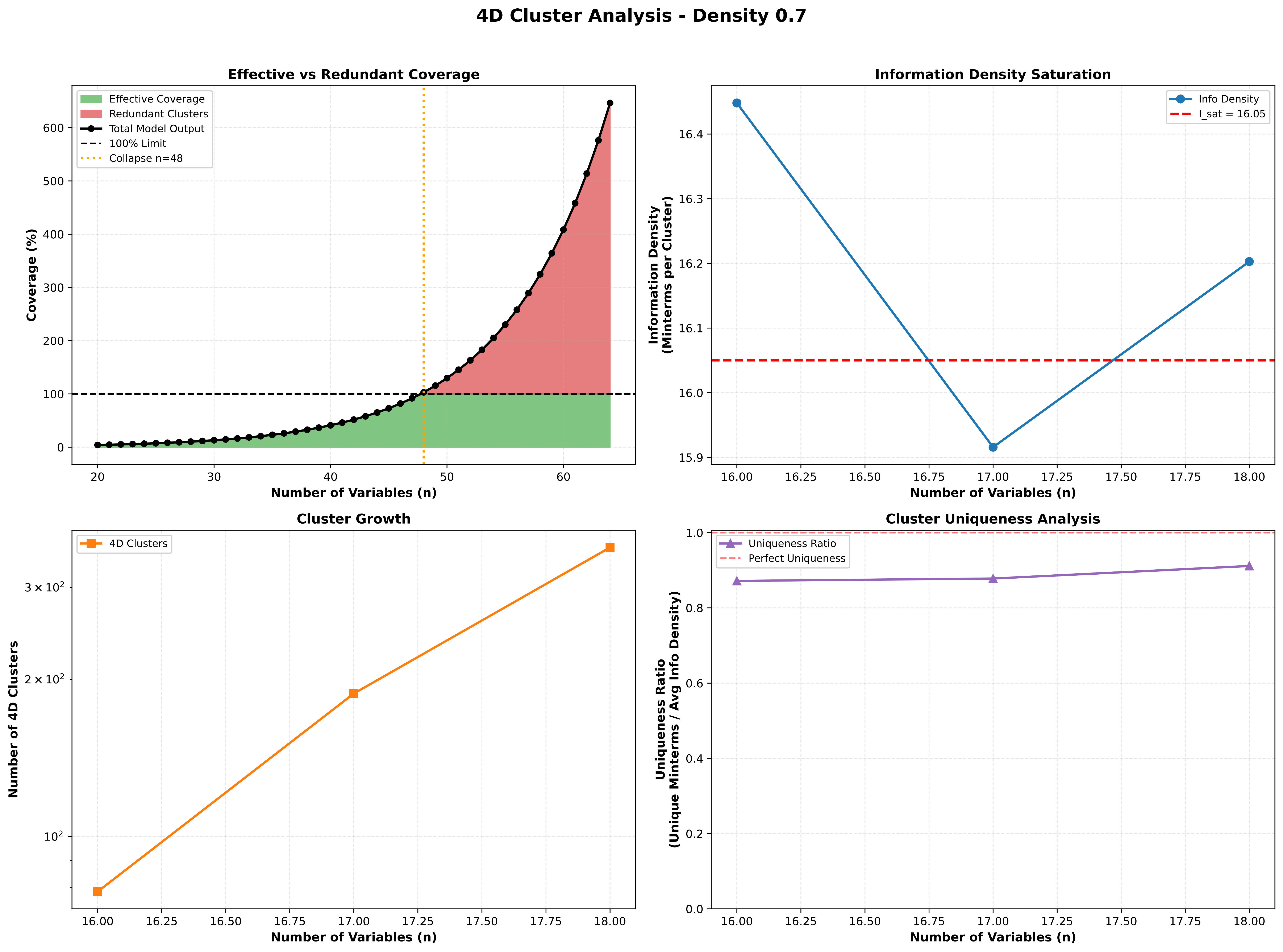
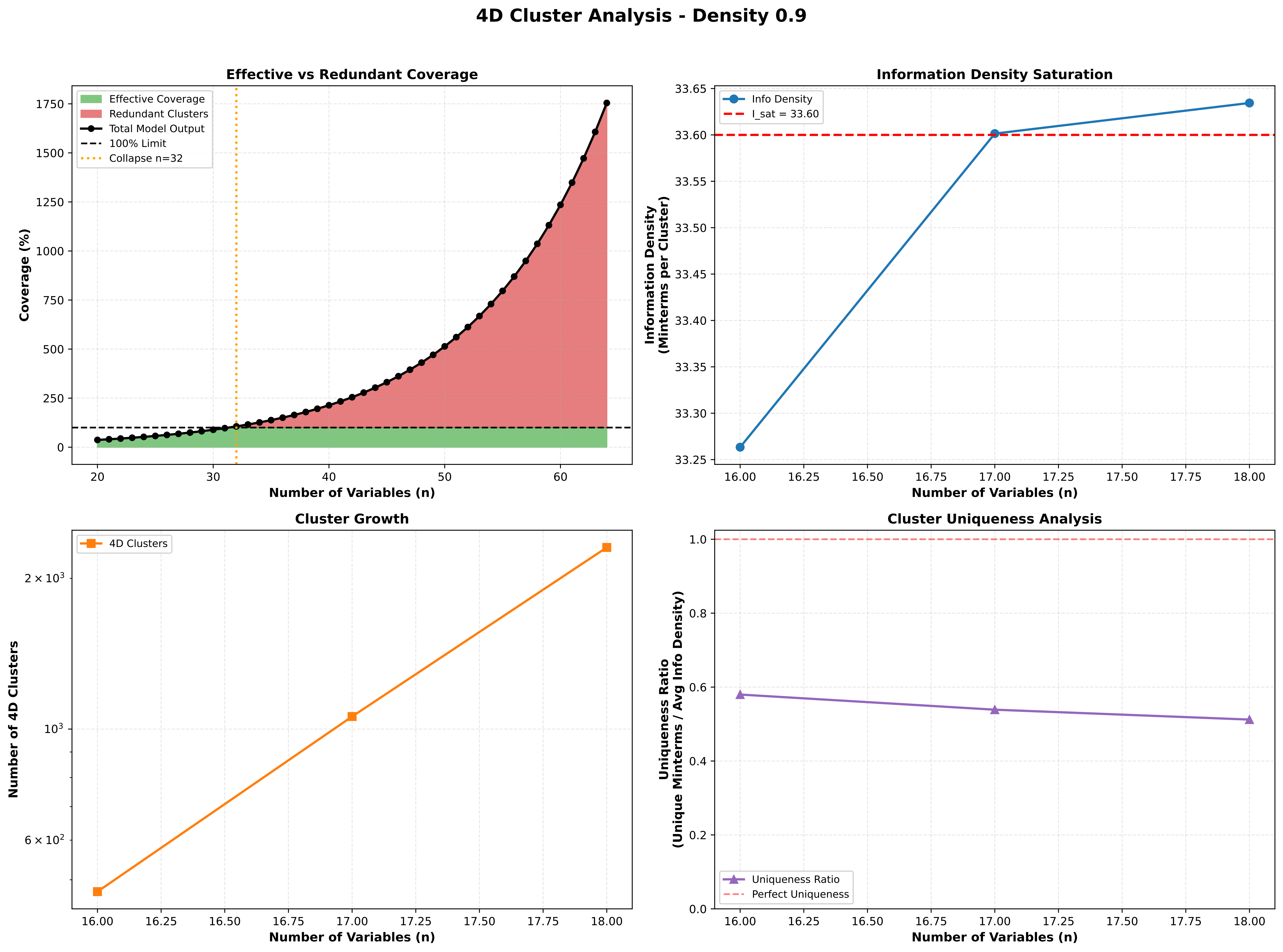
# 4D Cluster Analysis - Density 0.3





4D Cluster Analysis - Density 0.5
Effective vs Redundant Coverage
Effective Coverage
Redundant Clusters
Total Model Output
100% Limit
Collapse n=53
Coverage (%)
Number of Variables (n)
Information Density Saturation
Information Density
(Minterms per Cluster)
Info Density
I_sat = 9.05
Number of Variables (n)
Cluster Growth
4D Clusters
Number of 4D Clusters
Number of Variables (n)
Cluster Uniqueness Analysis
Uniqueness Ratio
(Unique Minterms / Avg Info Density)
Uniqueness Ratio
Perfect Uniqueness
Number of Variables (n)


4D Cluster Analysis - Density 0.7
Effective vs Redundant Coverage
Effective Coverage
Redundant Clusters
Total Model Output
100% Limit
Collapse n=48
Coverage (%)
Number of Variables (n)
Information Density Saturation
Info Density
I_sat = 16.05
Information Density (Minterms per Cluster)
Number of Variables (n)
Cluster Growth
4D Clusters
Number of 4D Clusters
Number of Variables (n)
Cluster Uniqueness Analysis
Uniqueness Ratio
Perfect Uniqueness
Uniqueness Ratio (Unique Minterms / Avg Info Density)
Number of Variables (n)


4D Cluster Analysis - Density 0.9
Effective vs Redundant Coverage
Effective Coverage
Redundant Clusters
Total Model Output
100% Limit
Collapse n=32
Coverage (%)
Number of Variables (n)
Information Density Saturation
Info Density
I_sat = 33.60
Information Density (Minterms per Cluster)
Number of Variables (n)
Cluster Growth
4D Clusters
Number of 4D Clusters
Number of Variables (n)
Cluster Uniqueness Analysis
Uniqueness Ratio
Perfect Uniqueness
Uniqueness Ratio (Unique Minterms / Avg Info Density)
Number of Variables (n)